Performance Testing

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

This report records the results of the automated test suites and the measurable outcomes observed during the technical audit of October 19, 2026. Only results actually recorded in the audit are reported as measured; items that were not benchmarked are listed separately so they are not mistaken for results.

1. Test Environment

Item	Value
Platform	macOS (Apple Silicon / Intel); also targets Linux x86_64 and Windows
Python	3.11.12 (CPython)
Key libraries	OpenCV 4.11.0, MediaPipe 0.10.35, DeepFace 0.0.102, DeepSort Realtime 1.3.2, Librosa 0.10.2, FastAPI 0.143.3, SQLAlchemy 2.1.5, Streamlit 1.66.0
Camera	Hardware camera 0 verified at 1280x720 by verify_env.py; pipeline works on 640x480 frames
Test media	Synthetic test_classroom.mp4 (640x480) and test_classroom.wav generated by generate_test_media.py
Location of tests	tests/ directory, 12 standalone scripts

2. Overall Result

12 / 12	0	100%
suites passed	suites failed	success rate

3. Test Suite Results

Test suite	Component	What was verified	Result
test_api_endpoints.py	FastAPI, WebSocket	Health check, session lifecycle, audio chunk ingestion, /ws/telemetry streaming, attendance correction, camera toggle	PASS
test_attendance_service.py	Attendance	Session duration, presence accumulation, PRESENT / PARTIAL / ABSENT thresholds, CSV export	PASS
test_audio_analysis.py	Audio features	Silence handling, 16-bit PCM decoding, RMS, dBFS, spectral centroid, MFCC on test_classroom.wav (2 s chunks)	PASS
test_database.py	SQLite / ORM	Creation of 5 tables; student, session, attendance, recommendation CRUD	PASS
test_end_to_end_pipeline.py	Full system	11 steps: health, enrollment, session start, audio, vision, roster, analytics, recommendations, WebSocket, finalize, CSV and DB check	PASS
test_engagement_engine.py	Engagement	Composite formula, attentive vs distracted scoring, EMA smoothing, class aggregation	PASS
test_face_detection.py	Detection, tracking	Haar detection, appearance embedding, DeepSort persistence on synthetic video	PASS
test_face_recognition.py	Recognition	Consent rejection, Facenet embedding, cosine matching below 0.15, unknown-face fallback, index reload	PASS
test_head_pose_gaze.py	Pose and gaze	478 landmark init, solvePnP yaw/pitch/roll, EAR, bounding-box fallback, annotation	PASS
test_multimodal_alignment.py	Fusion	Circular buffer fallback, async audio/video timestamp matching, multi-student sync, rolling stats	PASS
test_reasoning_engine.py	Recommendations	Warning vs urgent rules, 30 s debounce, acoustic rule, 100-item ring buffer, DB persistence	PASS
test_vad_noise_classifier.py	VAD / noise	Adaptive noise floor, speech detection, burst disturbance detection, 10 s sequence	PASS

4. Measured Outcomes From Tests

Area	Observation recorded	Source suite
Engagement scoring	Attentive student 0.968, distracted student 0.598, low-engagement student 0.224 (clear separation between profiles)	test_engagement_engine
Pose and gaze	69 face poses processed; mean attentiveness 0.72	test_head_pose_gaze
Tracking persistence	DeepSort track persistence of 66% and 98% on the two synthetic tracks	test_face_detection
Recognition	Enrolled faces matched at cosine distance below 0.15	test_face_recognition
Attendance	33.3% attendance rate across the test subjects in the attendance service test	test_attendance_service
Camera	Camera 0 verified at 1280x720; fallback to test_classroom.mp4 verified	verify_env.py
Database	attendance_records held 59 rows after use; person_metrics held 0 rows (by design)	Audit inspection

5. Design Performance Targets

Parameter	Target / configured value
Video stream	About 25-30 fps MJPEG from /api/video/feed
Telemetry	WebSocket broadcast at 4 Hz
Dashboard refresh	Default 3 s polling; selectable 1, 2, 3, 5, or 10 s
Audio chunk	2.0 s, 16 kHz mono (32,000 samples)
Sync buffer	60-second circular window
Recommendation cooldown	30 s per rule

These are the design and configuration values from the implementation. They are targets, not independently benchmarked measurements.

6. Not Yet Benchmarked (Recommended Next Tests)

Metric	Why it matters	Suggested method
End-to-end frame latency	Confirms real-time behaviour on the target laptop	Time stamps at capture and at annotated output over 5 min
Sustained fps vs number of faces	Shows how many students one camera can handle	Run with 1, 5, 10, 20 faces and log fps
CPU and memory use	Checks suitability for low-end laptops	Monitor with system tools during a 30-minute session
Recognition accuracy	Validates the 0.15 threshold under real lighting	Enroll N students; record true/false matches by lighting and angle
Gaze/pose agreement	Checks engagement against human labels	Compare scores with manual attention ratings on recorded test clips

7. Known Limitations Found During Testing

- **person_metrics unpopulated:** per-frame telemetry is not persisted, so detailed history cannot be replotted after a restart.
- **No background microphone listener:** audio must be posted to /api/audio/chunk; otherwise audio defaults to QUIET.
- **Threshold sensitivity:** the 0.15 cosine threshold is conservative; poor lighting may make an enrolled student appear anonymous.
- **Single camera coverage:** one webcam cannot cover a very large lecture hall.

8. Conclusion

All 12 test suites passed, covering each service and the full 11-step pipeline. The system behaves as designed on synthetic and live camera input. Real-world performance numbers (latency, load, recognition accuracy) are the main gap and are listed above as the next testing steps.